
EM 652: Electrochemical Energy Storage

Course Code	EM 652
Course Name	Electrochemical Energy Storage
L-T-P-C	3-0-0-3
Intended for	M.Tech Energy Engineering
Distribution	Specialisation elective
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Course Modules

- **Module 1:** Introduction, General Concepts, Idea of Thermal Energy Storage, Phase Change Materials, Reversible Chemical Reactions, Energy Storage in Organic Fuels (14)
- **Module 2:** Mechanical Energy Storage, Electromagnetic Energy Storage, Electrochemical Energy Storage, Lead Acid Batteries, Principles Determining the Voltages and Capacities of Electrochemical cells, Binary and Ternary Electrodes Under Equilibrium or Near Equilibrium Conditions, Insertion Reaction Electrodes, Electrode Reactions (14)
- **Module 3:** Primary, Non rechargeable Batteries, Negative and Positive Electrodes in other Rechargeable Aqueous Systems, The Zinc Electrodes in Aqueous Systems, The “Cadmium” Electrode, Metal Hydride Electrodes, Manganese Dioxide Electrodes in Aqueous Systems, The “Nickel” Electrode, Cause of the Memory Effect in “Nickel” Electrodes, Negative and Positive Electrodes in Lithium Systems (14)

Text Books

1. R. Huggins, *Energy Storage*, Springer, 2010